Editing the Code of Life: Plasmid Design and Cloning

How scientists cut, paste and copy DNA, and move it into bacteria

GOALS FOR TODAY

- Explain why we put genes into **plasmids**, and name the parts of a plasmid.
- Describe the main steps of **molecular cloning**.
- Explain how **restriction enzymes** and **ligase** cut and paste DNA.
- Explain how bacteria that took up the plasmid are **selected** and how **blue–white screening** works.

1. The big picture

Genetic engineering is done by molecular cloning: a set of methods for putting a piece of DNA of interest (the insert) into a carrier called a vector, which copies it inside a host organism. To make a bacterium produce a useful protein, we place the gene into a circular piece of DNA called a plasmid and then put the plasmid into the bacterium. The bacterium then reads the gene and makes the protein.

STARTER QUESTION

Name some round objects. Can DNA be round? (Yes — bacterial chromosomes and plasmids are circles.)

2. Plasmids

Plasmids are small, circular, double-stranded DNA molecules that copy themselves independently of the cell's main chromosome. They occur mainly in bacteria, and also naturally in archaea and in some eukaryotes such as yeast. Not every bacterium has one. Plasmids range from about 1 to over 1,000 kilo-base-pairs and usually carry from a few to about a hundred genes.

In nature plasmids give their host extra abilities — resistance to antibiotics, the ability to digest unusual foods, or the power to cause disease (virulence). Bacteria can pass plasmids to each other, which is one reason antibiotic resistance spreads. Well-known examples are pBR322, pUC18 and the Ti plasmid of Agrobacterium (used to put genes into plants).

In the lab plasmids are mostly artificial, designed to carry foreign DNA into a cell. Because they are made by people, they are called vectors or constructs. Plasmids are easy to modify and copy themselves in the cell, which makes them the favourite tool of bioengineers.

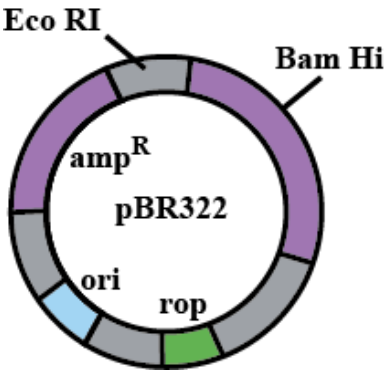


Figure 1. A simplified map of the plasmid pBR322, showing the antibiotic-resistance gene (ampR), the origin of replication (ori) and the gene rop. EcoRI and BamHI mark where restriction enzymes cut. (The real pBR322 also carries a tetracycline-resistance gene.) *Source: Day 3 session material*

Parts of a plasmid

Part	What it does
Origin of replication (ori)	The place where copying of the plasmid starts. It decides how many copies the cell holds and which hosts the plasmid can live in.
Selectable marker	Usually an antibiotic-resistance gene (for example ampicillin). Only cells that contain the plasmid survive on antibiotic plates.
Multiple cloning site (MCS) or polylinker	A short region with many restriction sites packed together, where the gene of interest is inserted.
Promoter	Switches on transcription of the inserted gene. In expression vectors it decides in which cells, and how strongly, the protein is made.
Optional extras	A ribosome-binding site, a terminator, a reporter gene (such as GFP) and tags that help purify the protein.

A minimal lab plasmid needs only three things: an origin of replication, a selectable marker, and a cloning site.

3. Molecular cloning step by step

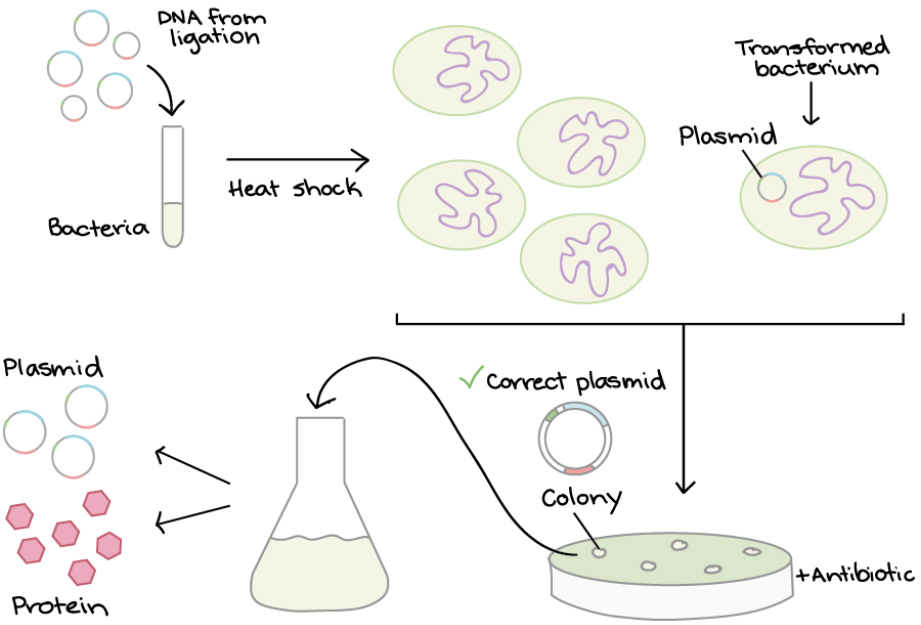


Figure 2. The cloning workflow: DNA from a ligation is mixed with bacteria, the bacteria get a heat shock, are grown on antibiotic plates, a colony with the right plasmid is picked and grown in bulk to make plasmid or protein. *Source: Khan Academy*

1. **Get the gene of interest.** Copy it from a genome by PCR, or (more commonly) order the synthetic gene from a company. This means we can skip extracting DNA from cells.
2. **Prepare the plasmid.** Grow bacteria that carry the plasmid and purify it using a plasmid-purification ("miniprep") kit, which separates plasmid DNA from the bacterium's chromosomal DNA.
3. **Cut** the insert and the plasmid with the **same restriction enzymes**.
4. **Paste (ligate)** the insert into the plasmid with **DNA ligase**.
5. **Transform** the new plasmid into bacteria.
6. **Select and screen:** grow on antibiotic plates; use blue–white screening to spot colonies with an insert.
7. **Verify** that the plasmid is correct.

WHERE ARE PCR AND GEL ELECTROPHORESIS?
Step 1 often uses PCR, and gels are used to check DNA fragments. Both are taught in the Viral Diagnosis session.

Step 3: Restriction enzymes — molecular scissors

A DNA molecule or genome is very long, so we cut it into small pieces. Restriction enzymes are natural bacterial proteins that cut DNA at a specific short sequence, the recognition (restriction) site. (Bacteria use them to chop up the DNA of invading viruses.) Most recognition sites are palindromic: the sequence on one strand read 5′ → 3′ is the same as the sequence on the other strand read 5′ → 3′.

Example: EcoRI recognises GAATTC and cuts between G and A on both strands:

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5′ ...G ▾AATTC... 3′
3′ ...CTTAA▴ G... 5′

after cutting:
5′ ...G          AATTC... 3′
3′ ...CTTAA          G... 5′
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The cut is staggered, so each fragment has a short single-stranded overhang (here AATT). These are sticky ends: they can base-pair with any other EcoRI-cut end. Because we cut both the insert and the plasmid with the same enzyme, their sticky ends match.

Enzyme	Recognition site (5′ → 3′)	Type of end
EcoRI	G^AATTC	Sticky (5′ overhang)
BamHI	G^GATCC	Sticky (5′ overhang)
HindIII	A^AGCTT	Sticky (5′ overhang)
SmaI	CCC^GGG	Blunt (no overhang)

Step 4: Ligation — molecular glue

When the sticky ends of the insert and the plasmid meet, they pair by hydrogen bonds. But the sugar–phosphate backbone still has gaps (nicks), so the join is weak. DNA ligase forms the missing phosphodiester bonds and seals the nicks, giving one continuous recombinant plasmid.

Step 5: Getting DNA into cells

Method	What it means	Used for
Transformation	A cell takes up naked DNA from its surroundings.	Bacteria, yeast (the lab method for plasmids)
Transduction	A virus carries DNA into the cell. In bacteria the virus is a bacteriophage; in animal cells, engineered viruses can deliver genes that then either stay free or integrate into the genome.	Bacteria in nature; gene therapy
Transfection	Nucleic acids are forced into animal cells by chemical, physical or biological methods. The DNA may stay temporarily (transient) or become permanent (stable).	Animal cell culture
Conjugation	Two bacteria connect and one copies a plasmid to the other.	Natural spread of plasmids

How bacterial transformation is done

For a cell to take up DNA it must first become competent (permeable to DNA). In nature some bacteria become competent under stress. In the lab we make cells competent on purpose by treating them with ice-cold calcium chloride (or rubidium or magnesium salts) and keeping them cold. Competent cells are fragile, so handle them gently and keep

them on ice. Competent cells differ in how well they take up DNA, which we measure as transformation efficiency.

Step	What to do	Why
1	Mix competent *E. coli* (kept on ice) with the plasmid.	Cold, calcium-treated cells bind DNA to their surface.
2	Keep on ice for about 30 minutes.	Lets DNA stick to the cell.
3	Heat shock: 42 °C for 30–45 seconds, then back on ice for 2 minutes.	The sudden change opens pores so DNA enters.
4	Add recovery broth and incubate at 37 °C for 30–60 minutes.	Cells repair their walls and start making the resistance protein.
5	Spread on an LB agar plate with the antibiotic.	Only transformed cells can grow.
6	Incubate at 37 °C overnight (16–18 hours).	Each survivor grows into a visible colony.

ALWAYS RUN CONTROLS

Prepare two tubes of competent cells: one with plasmid DNA and one without. The no-DNA tube is the negative control: it should give no colonies on antibiotic plates. If it does, something is contaminated. A tube with a plasmid you know works is a positive control: if it gives no colonies, the competent cells or the method failed.

CALCULATE: TRANSFORMATION EFFICIENCY

Efficiency = number of colonies ÷ amount of DNA used (in μg), if all the cells are plated.

Example: 1 ng of plasmid (0.001 μg) gives 300 colonies → 300 ÷ 0.001 = 300,000 colonies per μg (3×10^5). Too little DNA gives few colonies, but too much DNA also reduces efficiency.

4. Selecting the right bacteria

The host strain used for cloning (for example E. coli DH5α) is sensitive to all common antibiotics. If we add ampicillin to the agar, the only cells that can grow are those that took up a plasmid carrying the ampicillin-resistance gene. Cells without a plasmid are killed or stopped from dividing. Each surviving cell divides again and again to form a colony.

Blue–white screening

Not every colony has our insert. Some plasmids close up again without it (an empty vector). Blue–white screening shows which is which. The plasmid carries a short piece of the lacZ gene (from the lac operon you met on Day 2), and the multiple cloning site sits

inside it. The plates contain X-gal (a colourless substance) and IPTG (which switches the gene on).

Cell contains…	lacZ gene	On X-gal + IPTG + ampicillin
No plasmid	—	**No growth**
Empty plasmid (no insert)	Intact → working β-galactosidase	**Blue colonies**
Recombinant plasmid (with insert)	Interrupted by the insert → no working enzyme	**White colonies**

So the cloner picks white colonies. A white colony is a good candidate, but not proof, so the plasmid is always checked.

Verifying the clone

- **Colony PCR** or **restriction digest** and a gel, to check the insert is present and the right size.
- **DNA sequencing**, to confirm the exact sequence.
- Then grow the right colony in bulk, and purify the plasmid or the protein.

BIOSAFETY
Cloning strains such as DH5α are weakened lab strains and are handled at the lowest biosafety level (BSL-1), but they still carry antibiotic resistance. Never pour cultures down the sink. Disinfect or autoclave all plates, tubes and tips. See the Safety Guide.

Check your understanding

1. Name the three parts a lab plasmid must have.
2. Why do we cut the insert and the plasmid with the *same* restriction enzyme?
3. What does DNA ligase do that hydrogen bonding between sticky ends cannot?
4. What is the purpose of a no-DNA control in a transformation?
5. On a plate with ampicillin, X-gal and IPTG you see blue colonies, white colonies, and empty spaces. What does each mean?
6. Why can a white colony still contain the wrong plasmid?

Answers are in the Quizzes and Answer Keys document.